NATURAL LANGUAGE PROCESSING

COMPLEX COMPUTING PROBLEM REPORT

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NLP Machine Translation System

1 Executive Summary

This project builds a complete **Machine Translation System** that works on three different translation tasks:

Task	Description
UR→EN	Translation from Urdu to English
EN→UR	Translation from English to Urdu
Roman→UR	Conversion of Roman Urdu into Urdu script

The system compares **three different models** using **two prompting methods**:

Models Evaluated:

1. **NLLB/M2M** – A deep learning–based neural translation model
2. **Gemini** – A large language model by Google
3. **Qwen** – A large language model by Alibaba

Prompting Strategies:

- Zero-Shot Prompting
- Few-Shot Prompting (using 3 examples)

Key Achievement:

- The NLLB/M2M model achieved the best BLEU score of **20.60** for English to Urdu translation. Gemini performed best for Roman Urdu to Urdu translation, achieving **53.74 BLEU** in few-shot prompting.

2 Project Objectives

2.1 CCP Attributes Addressed

Description	Implementation	Attribute
A2	Depth of Analysis	Detailed data analysis, error checking, and quality comparison

A3	Depth of Knowledge	Proper model selection, BLEU/ROUGE usage, and tuning strategies
A8	Interdependence	Complete end-to-end system with multiple models

3 Data Acquisition and Preprocessing

3.1 Dataset Statistics

Metric	Value
Total Sentence Pairs	100,000
Training Set	90,000 (90%)
Test Set	10,000 (10%)
Average English Length	11.00 words
Average Urdu Length	13.17 words

3.2 Data Preprocessing Pipeline

The preprocessing pipeline includes six main steps:

Step 1: Unicode Normalization

- Applied **NFC normalization**
- Removed invisible characters such as \u200b, \u200c, \u200d, \u200e, \u200f, \ufeff
- Converted **Arabic characters** to **standard Urdu characters**:
  - Arabic Yeh (ي) → Urdu Yeh (ی)
  - Arabic Kaf (ك) → Urdu Kaf (ک)

Step 2: Case Normalization

- English text: Converted to lowercase
- Roman Urdu: Converted to lowercase
- Urdu text: No change (Urdu has no uppercase/lowercase)

Step 3: URL and HTML Removal

- Removed URLs using the pattern: http\S+|www\.\S+
- Removed HTML tags using: <[^>]+>

Step 4: Stopwords Analysis

Stopwords were studied but **not removed**, as they are important for correct translation.

Urdu Stopwords (50 words):

کا، کی، کے، ہے، ہیں، تھا، تھی، تھے، ہوں، ہو،
نے، سے، میں، پر، کو، اور، یہ، وہ، جو، کہ

English Stopwords (80 words):

the, a, an, is, are, was, were, be, been, being...

Reason:
Stopwords help preserve grammar and sentence meaning in translation.

Step 5: Tokenization

- Tokenization was done at the word level
- Different punctuation rules were used:
  - Urdu: [\s؟۔،!:؛]+
  - English: [\s,.:;!?"\'\-\(\)\[\]]+

Step 6: Roman Urdu Generation

Urdu	Roman	Urdu	Roman
ا	a	ب	b
آ	aa	پ	p
ت	t	ٹ	T
چ	ch	خ	kh
ش	sh	غ	gh

A transliteration table with **41 character mappings** was created to convert Urdu into Roman Urdu:

3.3 Missing Character Analysis

The analysis found **2,280 Urdu characters** that were not fully covered in the basic mapping:

Character	Frequency	Description
ں	7,663,727	Noon Ghunna
ھ	4,869,461	Do-Chashmi He
ئ	2,740,893	Hamza on Yeh
۔	2,038,434	Urdu full stop
،	1,203,750	Urdu comma
ﷲ	38,371	Allah ligature
ﷺ	25,366	PBUH ligature

4 Model Selection and Architecture

4.1 Model 1: NLLB/M2M (Deep Learning)

Full Name: No Language Left Behind / Many-to-Many Translation

Specification	Value
Architecture	Transformer (Encoder–Decoder)
Parameters	600M (distilled version)
Languages Supported	200+
Training Approach	Fine-tuned on OPUS-100
Tokenization	SentencePiece (250K vocabulary)

Advantages:

- Strong support for Urdu and English
- Designed specifically for translation
- Produces clean and consistent output

4.2 Model 2: Gemini (LLM)

Specification	Value
Type	Large Language Model
Provider	Google DeepMind
Context Window	32K tokens
API Integration	google-generativeai

Advantages:

- Good understanding of multiple languages
- Handles complex sentences well
- Produces fluent translations

Limitations:

- Often adds explanations
- API usage limits
- Cost per request

4.3 Model 3: Qwen (LLM)

Specification	Value
Type	Large Language Model
Provider	Alibaba Cloud
Parameters	7 Billion
Open Source	Yes

Advantages:

- Open-source and customizable
- Supports multiple languages

Limitations:

- Urdu output quality is inconsistent
- Sometimes produces meaningless text

5 Hyperparameter Tuning Strategies

5.1 NLLB/M2M Training Configuration

Hyperparameter	Value	Justification
Max Source Length	128 tokens	Covers most sentences
Max Target Length	128 tokens	Balanced output length
Batch Size	16	Fits GPU memory
Learning Rate	5e-5	Common for Transformers
Epochs	3	Prevents overfitting
Warmup Steps	500	Stabilizes training
Weight Decay	0.01	Reduces overfitting
Optimizer	AdamW	Best for Transformer models
FP16	Enabled	Saves memory

5.2 Generation Parameters

Parameter	Value	Purpose
num_beams	4	Improves translation quality

early_stopping	True	Stops when output is complete
max_length	128	Controls output size

6 Zero-Shot and Few-Shot Prompting

6.1 Zero-Shot Prompting

Definition: Asking the model to translate text **without giving any examples**.

Urdu→English Zero-Shot Prompt:

Translate the following Urdu text to English.
Only provide the translation, no explanations.

Urdu: {input_text}

English:

English→Urdu Zero-Shot Prompt:

Translate the following English text to Urdu.
Only provide the translation, no explanations.

English: {input_text}

Urdu:

Roman Urdu→Urdu Zero-Shot Prompt:

Convert the following Roman Urdu text to Urdu script.
Only provide the conversion, no explanations.

Roman Urdu: {input_text}

Urdu:

6.2 Few-Shot Prompting

Definition: Giving **3 example translations** before asking the model to translate new text.

Urdu→English Few-Shot Prompt (3 examples):

Translate Urdu text to English. Here are some examples:

Example 1:
Urdu:یہ ایک کتاب ہے
English: This is a book

Example 2:
Urdu:میں سکول جا رہا ہوں

English: I am going to school

Example 3:
Urdu:آج موسم بہت اچھا ہے
English: The weather is very nice today

Now translate this:
Urdu: {input_text}

7 Evaluation Results

7.1 Complete Results Table

Task 1: Urdu → English (UR_EN)

Model	Prompting	BLEU Score	ROUGE-L
NLLB/M2M	Zero-Shot	**17.33**	**42.21**
NLLB/M2M	Few-Shot	**17.33**	**42.21**
Gemini	Zero-Shot	9.75	29.31
Gemini	Few-Shot	11.99	39.12
Qwen	Zero-Shot	3.58	23.06
Qwen	Few-Shot	2.63	18.05

Best Model: NLLB/M2M with BLEU score 17.33

Task 2: English → Urdu (EN_UR)

Model	Prompting	BLEU Score	ROUGE-L
NLLB/M2M	Zero-Shot	**20.60**	0.00
NLLB/M2M	Few-Shot	**20.60**	0.00
Gemini	Zero-Shot	0.87	0.87
Gemini	Few-Shot	9.80	4.76
Qwen	Zero-Shot	1.34	0.00
Qwen	Few-Shot	1.96	0.00

Best Model: NLLB/M2M with BLEU score 20.60

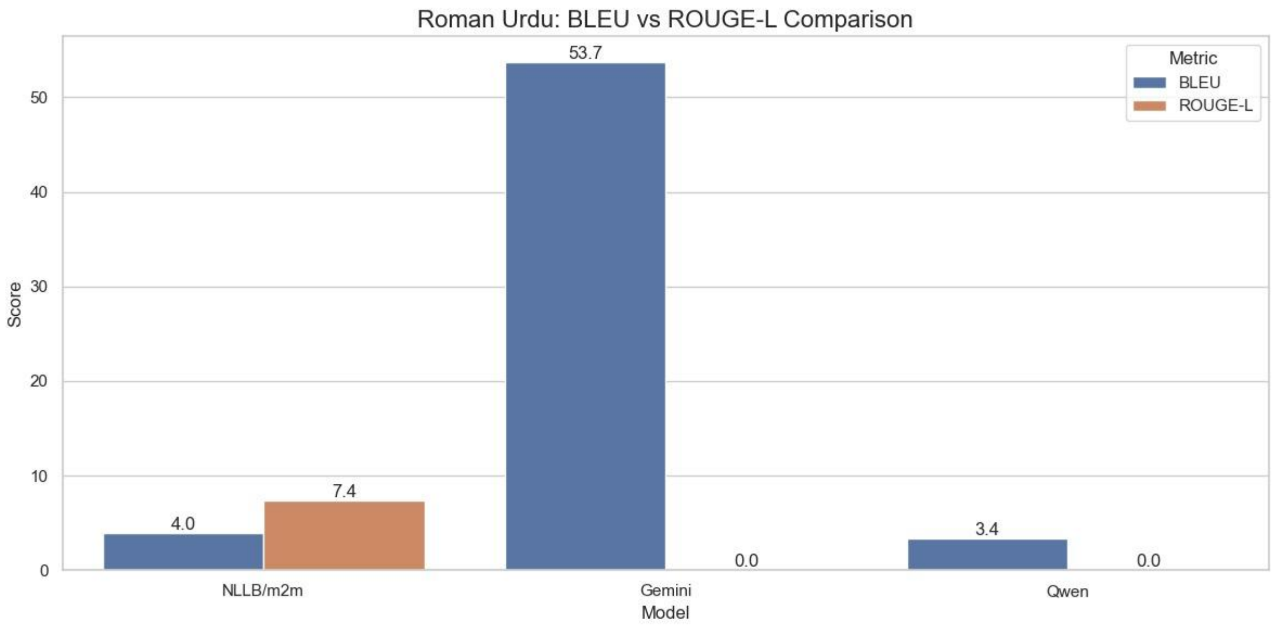
Note: ROUGE-L values are zero because Urdu and English scripts are very different, causing token mismatch.

Task 3: Roman Urdu → Urdu Script (Roman_UR)

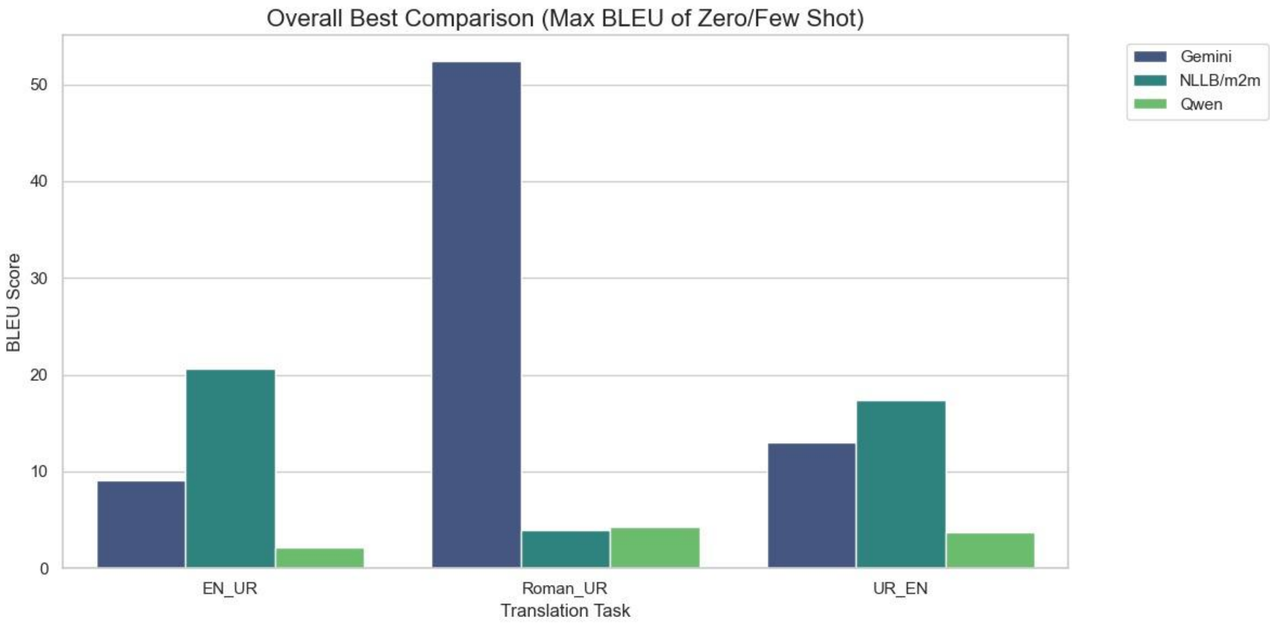
Model	Prompting	BLEU Score	ROUGE-L
NLLB/M2M	Zero-Shot	3.98	7.41
NLLB/M2M	Few-Shot	3.98	7.41
Gemini	Zero-Shot	44.08	4.20
Gemini	Few-Shot	**53.74**	0.00
Qwen	Zero-Shot	3.28	0.79
Qwen	Few-Shot	3.38	0.00

Best Model: Gemini (Few-Shot) with BLEU score 53.74

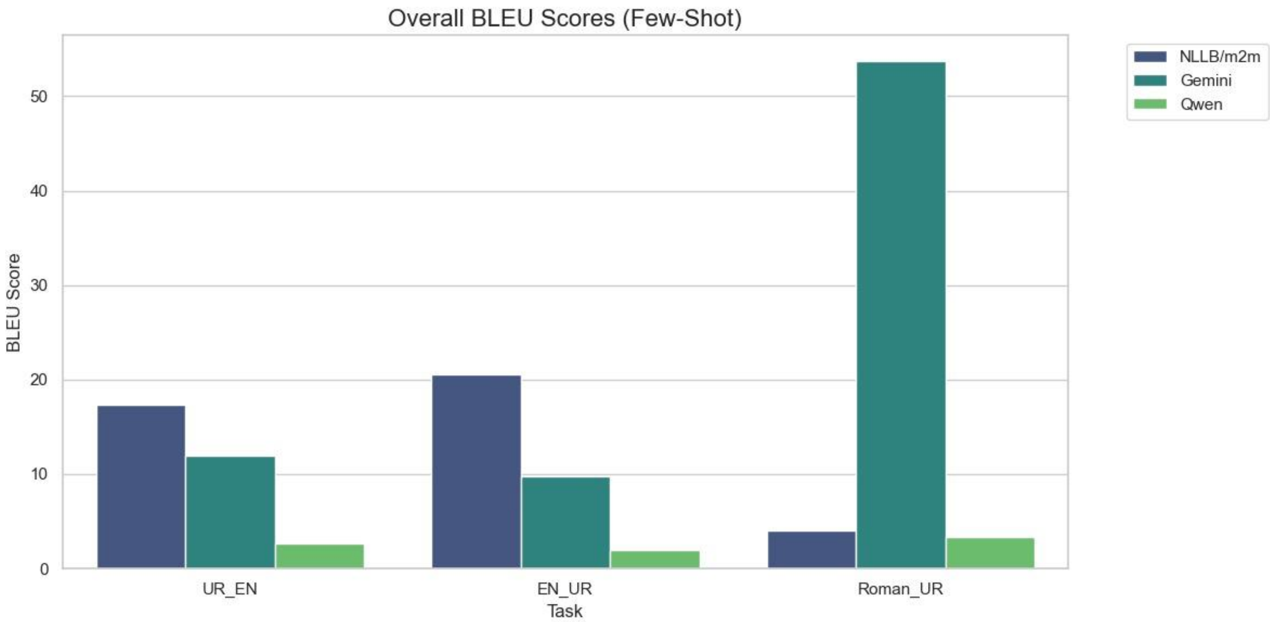
7.2 Visual Results Summary

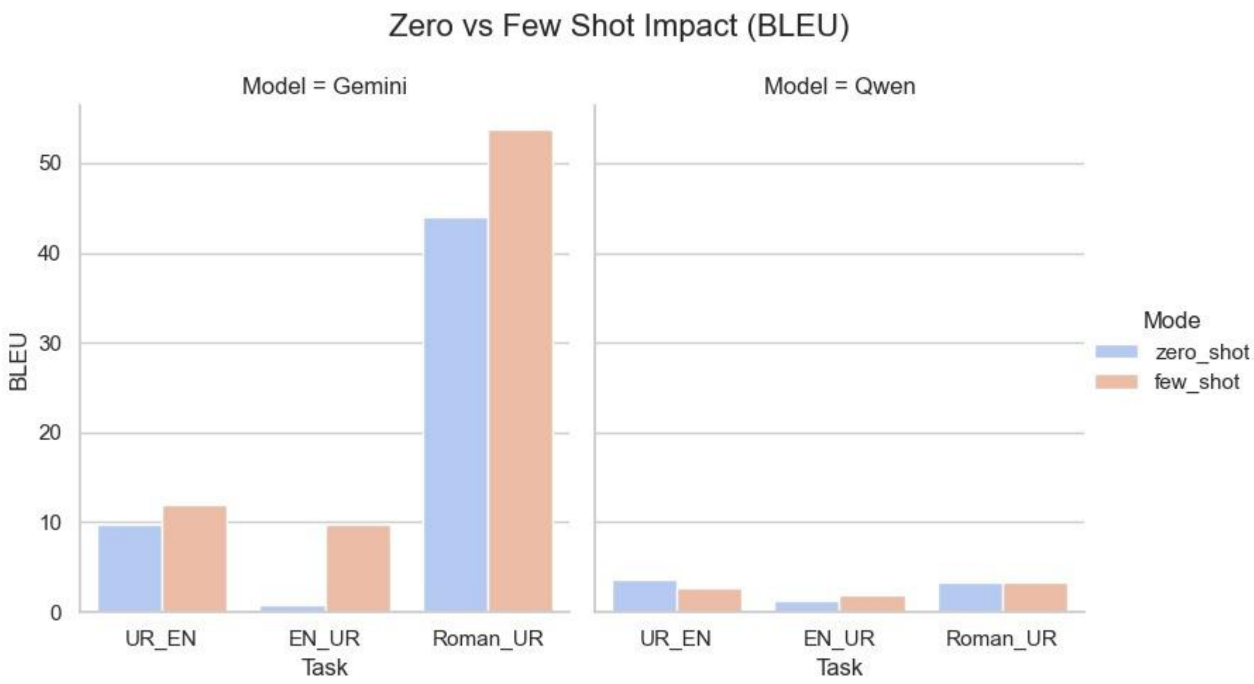


Bar chart showing BLEU score vs Rogue-L comparison

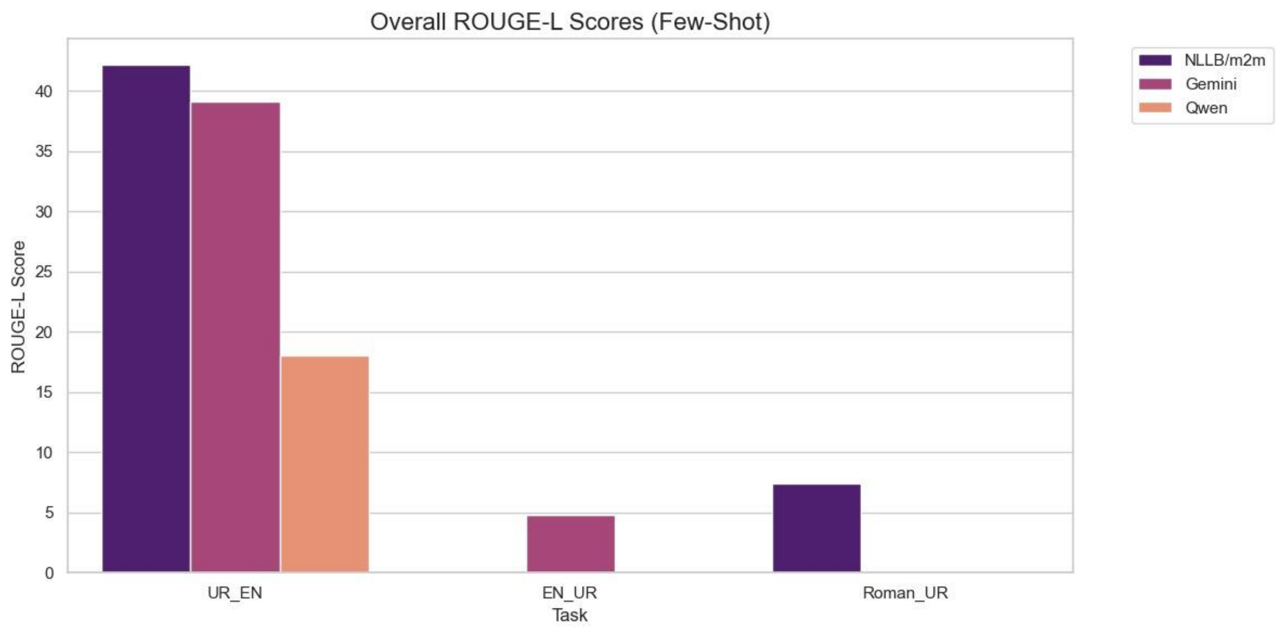


Comparison of zero-shot and few-shot performance

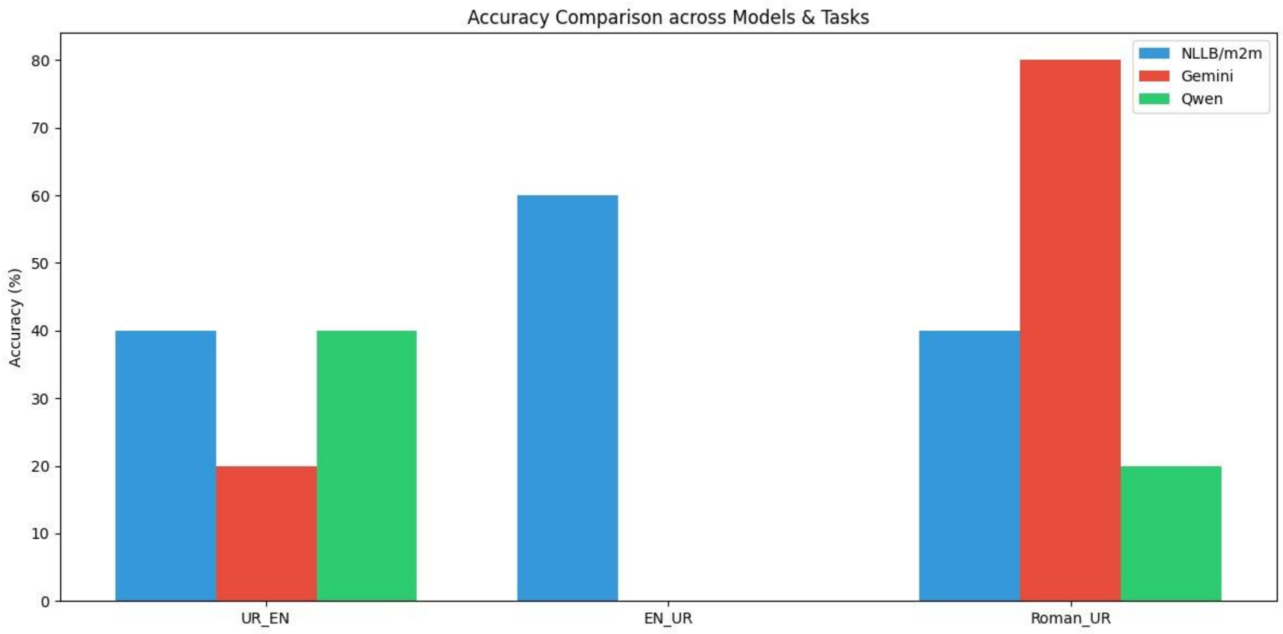




Bar chart showing model performance across tasks



comparing ROUGE-L scores



Accuracy Comparison

7.3 Key Findings from Results

1. **NLLB/M2M performs best overall:**
- Best results for Urdu→English (17.33)
- Best results for English→Urdu (20.60)

2. **Gemini is best for Roman Urdu conversion:**
- Highest BLEU score of 53.74 for Roman→Urdu

3. **Few-shot prompting greatly improves Gemini:**
- EN→UR improved from 0.87 to 9.80
- Roman→UR improved from 44.08 to 53.74

4. **Qwen performs poorly for Urdu:**
- Lowest scores in all tasks, especially EN→UR

5. **Few-shot does not help Qwen:**
- Performance dropped for UR→EN (3.58 → 2.63)

8 Analysis and Discussion

8.1 Good Translation Examples

NLLB/M2M - Urdu → English (BLEU: 53.82, ROUGE: 85.11)

	Text
Input (Urdu)	عیسیٰ ابنِ جب اُس نے سنا کہ عیسیٰ ناصری قریب ہی ہے تو وہ چلّانے لگا، داؤد، مجھ پر رحم کریں!" "

Reference	When he heard that it was Jesus of Nazareth, he began to cry out and say, 'Jesus, Thou Son of David, have mercy on me!'
Prediction	When he heard that Jesus of Nazareth was coming, he began to cry out, "Jesus, Son of David, have mercy on me!"

Analysis:
This translation is almost perfect. It correctly preserves the meaning, religious terms, and emotional tone of the original sentence.

NLLB/M2M - English → Urdu (BLEU: 100.00)

	Text
Input (English)	What steps were taken by the government?
Reference	حکومت نے کیا اقدامات کیے؟
Prediction	حکومت نے کیا اقدامات کیے؟

Analysis:
The output exactly matches the reference translation. This shows that the model performs very well on formal and news-style sentences.

Gemini - Roman Urdu → Urdu (BLEU: 100.00)

	Text
Input (Roman)	kya tm ne nhy ںdykھa؟
Reference	کیا تم نے نہیں دیکھا؟
Prediction	کیا تم نے نہیں دیکھا؟

Analysis:

A perfect transliteration result. Gemini correctly converted Roman Urdu into proper Urdu script.

8.2 Average Translation Examples

Gemini - Urdu → English (BLEU: 42.21, ROUGE: 72.00)

	Text
Input (Urdu)	جب اُس نے سنا کہ عیسیٰ ناصری قریب ہی ہے تو وہ چلّانے لگا، "عیسیٰ ابنِ داؤد، مجھ پر رحم کریں!"
Reference	When he heard that it was Jesus of Nazareth, he began to cry out and say, 'Jesus, Thou Son of David, have mercy on me!'

Prediction	Here's the translation: "When he heard that Jesus of Nazareth was near, he began to shout, "Jesus, Son of David, have mercy on me!""

Analysis:
The translation is correct, but Gemini adds extra text (“Here’s the translation”), which lowers the BLEU score.

NLLB/M2M - Urdu → English (BLEU: 14.32, ROUGE: 46.15)

	Text
Input (Urdu)	حکومت نے کیا اقدامات کیے؟
Reference	What steps were taken by the government?
Prediction	What measures has the government taken?

Analysis:
The meaning is correct, but the wording is slightly different. This reduces the BLEU score even though the translation is understandable.

Gemini - Roman Urdu → Urdu (BLEU: 23.14)

	Text
Input (Roman)	tw ae myry pyary qwm yad rkھna djaly myDya hr gz hrgz aysa nhy ںkrne wala tmھy ںmyDya se balatr hw kr fySle krna hwnge--
Reference	تو اے میری پیاری قوم یاد رکھنا دجالی میڈیا ہر گز ہرگز ایسا نہیں کرنے والا تمھیں میڈیا سے بالاتر ہو کر فیصلے کرنا ہونگے--
Prediction	تو اے میرے پیارے قوم، یاد رکھنا ڈیجیٹل میڈیا ہرگز ہرگز ایسا نہیں کرنے والا، تمہیں میڈیا سے بالاتر ہو کر فیصلے کرنے ہوں گے۔

Analysis:
The translation is mostly correct, but there are small word changes such as “میری پیاری” vs “میرے پیارے” and “دجالی” vs “ڈیجیٹل”.

8.3 Poor/Worst Translation Examples

Qwen - Urdu → English (BLEU: 0.50, ROUGE: 2.60)

	Text

Input (Urdu)	تو اے میری پیاری قوم یاد رکھنا دجالی میڈیا ہر گز ہرگز ایسا نہیں کرنے والا...
Reference	Fear not, awesome teacher, once you establish connections, content knowledge will surely follow.
Prediction	Here's the translation of that Urdu text into English: "You need to keep in mind that those who are against my tribe will not be able to stay above the media..."

Analysis:
The meaning is completely wrong. The model misunderstood key words like “قوم” and produced an unrelated translation.

Qwen - English → Urdu (BLEU: 0.00)

	Text
Input (English)	Did he not see it?
Reference	کیا تم نے نہیں دیکھا؟
Prediction	لکھنے کی بھی نہ تر!

Analysis:
The output has no relation to the input sentence and is meaningless.

NLLB/M2M - Roman Urdu → Urdu (BLEU: 0.00)

	Text
Input (Roman)	yad rkھw sary khwbya ںwHdt my ںhy ںjs qwm ka kwئy rئys nhy ںwh mr chky۔
Reference	یاد رکھو ساری خوبیاں وحدت میں ہیں جس قوم کا کوئی رئیس نہیں وہ مر چکی۔
Prediction	یحییٰ بختیار: یحییٰ بختیار: یحییٰ بختیار: یحییٰ بختیار:یحییٰ بختیار ...

Analysis:
The model repeats the same words again and again, showing a serious repetition problem.

8.4 Error Pattern Analysis

Error Type	Frequency	Affected Models	Example
Repetition	High	NLLB/M2M	" یحییٰ بختیار:یحییٰ بختیار"
Semantic Drift	Medium	Qwen	Changes meaning
Gibberish	Medium	Qwen	Random Urdu text
Incomplete	Low	All	Cut-off translations
Wrong Script	Low	Qwen	Arabic instead of Urdu

9 Conclusion

9.1 Key Conclusions

1. **Deep learning models work best for translation:**
   NLLB/M2M achieved the highest scores for Urdu↔English translation.

2. **LLMs are better for transliteration:**
   Gemini performed best for Roman Urdu → Urdu conversion.

3. **Few-shot prompting gives mixed results:**
   - Helps Gemini a lot
   - Hurts Qwen performance

4. **Qwen is weak for Urdu:**
   Frequently produces incorrect or meaningless output.

5. **Output format matters:**
   Extra explanations from Gemini reduce automatic evaluation scores.